How Large Is the Hyperbolic Advantage?

A strong-baseline audit for multimodal retrieval

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Reproducible deck: frozen recipes, five seeds, rank-based evidence

Take-away: useful on one regime, not universal

What survives the audit

- Direct Lorentz lift + full-training LoRA is the strongest H variant.
- Development: a local sweet spot at 128-D.
- Full-corpus NanoSciFact improves at both 128-D and 256-D.

What does not survive

- No broad win on the 97-task MMEB-v3 screen.
- Video/image retrieval does not reliably improve.
- Raw cosine/E/H margins are not on a shared absolute scale.

The result is **task-conditional evidence**, not a universal geometry claim.

Why hyperbolic space suggests larger capacity

For the unit-curvature Lorentz model,

$$\mathbb{H}^n = \{x : \langle x, x\rangle_L = -1,\ x_0 > 0\}, \qquad ds^2 = dR^2 + \sinh^2 R\, d\Omega_{n-1}^2.$$

Thus shell area grows as $\sinh^{n-1} R = \Theta(e^{(n-1)R})$.
For two equal-radius points separated by angle θ,

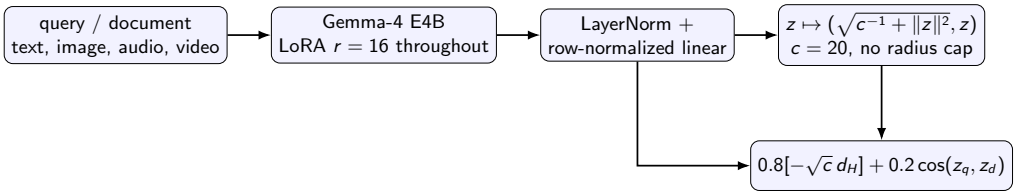
$$\cosh d_H = 1 + \sinh^2 R(1 - \cos\theta), \qquad d_H \approx 2R + 2\log\theta - \log 4.$$

If N directions require $\theta \asymp N^{-1/(n-1)}$, constant separation needs $R_H = O(\log N)$, versus polynomial Euclidean radius.

Scope

This is a representational-capacity argument under a shell-packing model. It does not guarantee that optimization, data, or downstream ranking realizes the capacity.

Final method: H10 is a hybrid Lorentz retriever



Symmetric hard-negative InfoNCE; 7 online negatives; batch 8×2 accumulation; 256 steps; no explicit margin loss. Cosine is computed *before* the lift.

A fair comparison requires a strong Euclidean baseline

Independent tuning

- 38 recipes $\times$ 3 seeds per geometry
- top 8 recipes $\times$ 2 new seeds
- same backbone, LoRA rank, training data, hard-negative count

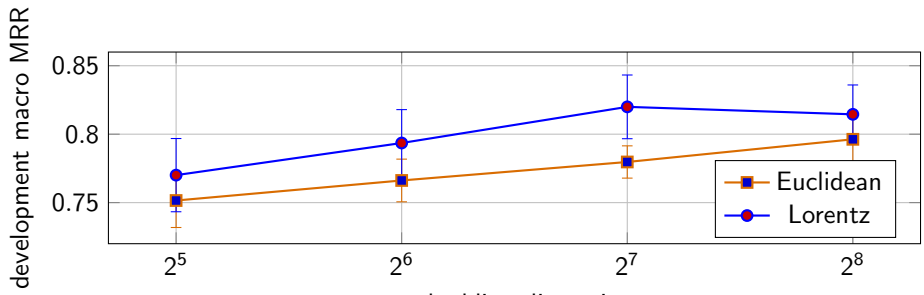
Matched-step Euclidean arm

- 256 steps / 16 warmup
- $R_{\max} = 2$, $\tau = 0.14$
- 0.2 pre-lift cosine mixture
- independent best head LR 2.5×10^{-4}

Interpretation rule

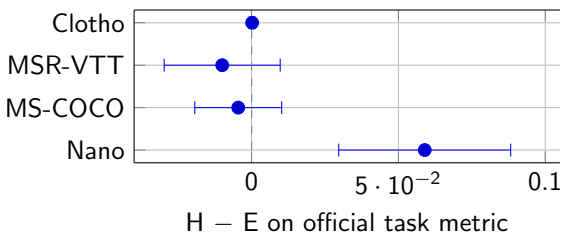
Systems differ in selected hyperparameters beyond geometry. This is a selected-system comparison, not a curvature-only causal intervention.

Development ablation: a local 128-D advantage



Five matched seeds on CIRR, ChartQA, WebQA. At 128-D: $H - E = +0.0403$ [95% CI $+0.0111, +0.0694$], wins 5/5. At 256, 64, and 32-D the CI crosses zero.

Frozen MMEB-v3 full-corpus confirmation



Task	E	H	$H - E$
NanoSciFact	0.2258	0.2847	+0.0590
MS-COCO T→I	0.0982	0.0936	-0.0045
MSR-VTT	0.0682	0.0582	-0.0100
Clotho	0.0006	0.0008	+0.0002

At 128-D, only NanoSciFact is positive with a non-crossing CI: +0.0590 nDCG@5, five wins in five seeds. Removing NanoSciFact, the other three-task macro delta is −0.0048 and crosses zero.

The broad screen is the necessary negative result

Dim	E	H	$H - E$ [95% CI]	W/T/L
128	0.2487	0.2447	-0.0040 [-0.0188, +0.0108]	40/2/55
256	0.2626	0.2596	-0.0030 [-0.0192, +0.0133]	41/3/53

128-D ranking transitions

- relevant rank: 11,183 improved / 5,524 tied / 12,228 worsened
- top-1: 1,558 became correct / 1,547 became wrong

Hard negatives

- 21,030 demoted / 691 promoted / 7,214 tied
- demotion alone does not imply better official metrics

Bounded pool: at most 64 queries and 1,024 candidates, with qrels used only to retain positives. This is a diagnostic screen, not an official full-corpus score.

Where does it work? Current best hypothesis

Supports H10

- NanoSciFact: text retrieval with scientific topical structure
- bounded MultiConIR People / Movies: positive at both dimensions (uncorrected)
- 128-D development sweet spot

Does not support H10

- MS-COCO text-to-image
- MSR-VTT video retrieval
- Clotho audio retrieval
- VisDoc family at 128-D

The emerging signal is **semantic/hierarchical text structure**, not omni-modality by itself.
This hypothesis requires a new pre-registered test.

Related work and what is different here

Line	Representative work	Difference in this audit
Multimodal embedding	CLIP; VLM2Vec; GME; MM-Embed	Strong MLLM embeddings and broad benchmarks, usually Euclidean/cosine
Hyperbolic representation	Poincaré embeddings; Lorentz embeddings	Hierarchy capacity and stable Lorentz optimization
Hyperbolic vision-language	MERU; HyCoCLIP; PHyCLIP; HALO	Direct lift and entailment/compositional objectives
This work	Gemma-4 + MMEB-v3	Independently tuned E/H systems, dimension ablation, broad negative result, frozen full-corpus confirmation

Key references: Nickel & Kiela (NeurIPS 2017; ICML 2018); Desai et al. (ICML 2023); Pal et al. (ICLR 2025); Huang et al. (MMEB-v3, 2026); Ratcliffe (*Foundations of Hyperbolic Manifolds*, 2019).

Conclusions and next falsifiable experiment

1. Hyperbolic volume growth gives a clean *capacity mechanism*, not an automatic ranking win.
2. H10 beats the matched Euclidean system at 128-D on development and on full-corpus NanoSciFact.
3. It does not improve the 97-task screen overall; multimodal transfer remains weak.
4. Cross-geometry raw distance margins are not evidence; rankings and official metrics are.

Next test

Pre-register a text-heavy hierarchy suite, tune neither geometry on its qrels, and compare 32/64/128/256-D with at least five new seeds. The hypothesis is that H helps when candidate structure is genuinely hierarchical, especially under moderate compression.

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